

Year 7 • Chapter 5B • Equations: answers (teacher copy)

5.10 Guess, check and improve (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: yes
- Ex 1b: $m = 4$
- Ex 2a: $k = 0$
- Ex 2b: $d = 18$
- Ex 3a: $a = 9$
- Ex 3b: $x = 0.5$

Easy • Set A (quick drill)

- | | | |
|--------|--------|--------|
| 1. yes | 5. no | 9. 14 |
| 2. no | 6. yes | 10. 4 |
| 3. yes | 7. 8 | 11. 9 |
| 4. yes | 8. 8 | 12. 15 |

Easy • Set B

- $= 7$
- $= 9$
- $= 17$
- $= 4$
- $= 4$
- $= 20$

Medium • Set C

- $m = 4$
- $k = 0$
- $d = 18$
- $a = 9$
- $b = 7$
- $x = 7$

Challenging • Set D

- $x = 0.5$
- $n = 12$ (or -12)
- $x = 5$ (or -6)
- 8

Skill drill 1 (Easy basics)

- | | | |
|---------|--------------|--------------|
| 1. no | 15. yes | 29. $x = 14$ |
| 2. yes | 16. no | 30. $m = 14$ |
| 3. no | 17. no | 31. $y = 4$ |
| 4. yes | 18. yes | 32. $t = 9$ |
| 5. yes | 19. no | 33. $k = 3$ |
| 6. yes | 20. yes | 34. $m = 14$ |
| 7. no | 21. no | 35. $a = 14$ |
| 8. no | 22. yes | 36. $b = 10$ |
| 9. no | 23. no | 37. $m = 12$ |
| 10. yes | 24. no | 38. $w = 10$ |
| 11. yes | 25. $p = 10$ | 39. $k = 15$ |
| 12. yes | 26. $b = 10$ | 40. $a = 8$ |
| 13. no | 27. $k = 16$ | |
| 14. yes | 28. $m = 1$ | |

Skill drill 2 (Medium basics)

- | | | |
|-------------|--------------|--------------|
| 1. $n = 15$ | 9. $b = 7$ | 17. $w = 5$ |
| 2. $a = 7$ | 10. $n = 7$ | 18. $k = 11$ |
| 3. $b = 14$ | 11. $p = 7$ | 19. $k = 3$ |
| 4. $y = 6$ | 12. $b = 5$ | 20. $b = 5$ |
| 5. $w = 12$ | 13. $a = 12$ | 21. $w = 6$ |
| 6. $b = 10$ | 14. $t = 4$ | 22. $k = 9$ |
| 7. $x = 12$ | 15. $w = 10$ | 23. $t = 8$ |
| 8. $y = 7$ | 16. $n = 5$ | 24. $w = 4$ |

Extension

- $x \approx 3.7$
- $x \approx 6.6$

Exit ticket

- Level 1: yes
- Level 2: $x = 7$
- Level 3: $y = 20$

5.11 Solving one-step equations (booklet p. 9)

WE DO examples (teacher)

- Ex 1a: $w = 35$
- Ex 1b: $e = 12$
- Ex 2a: $p = 3$
- Ex 2b: $x = 12$
- Ex 3a: $y = -7$
- Ex 3b: $k = -2.5$

Easy • Set A (quick drill)

- | | | |
|-------|---------|----------|
| 1. 35 | 6. 20 | 11. 7 |
| 2. 12 | 7. -5 | 12. -5 |
| 3. 11 | 8. 15 | 13. 21 |
| 4. 20 | 9. 0 | 14. -2 |
| 5. 15 | 10. 20 | |

Easy • Set B

- 3
- 12
- 9
- 45
- 5
- 30

Medium • Set C

- $y = -7$
- $k = -2.5$
- $m = -15$
- $x = -4$
- $b = 3.5$
- $q = 14$

Challenging • Set D

- $n + 19 = 45$, $n = 26$
- $7n = 84$, $n = 12$
- $n/4 = 11$, $n = 44$
- $n - 23 = 58$, $n = 81$

Skill drill 1 (Easy basics)

- | | | |
|--------------|--------------|--------------|
| 1. $k = 25$ | 15. $n = 32$ | 29. $a = 18$ |
| 2. $w = 17$ | 16. $b = 3$ | 30. $n = 33$ |
| 3. $w = 9$ | 17. $n = 6$ | 31. $k = 20$ |
| 4. $b = 26$ | 18. $k = 25$ | 32. $b = 6$ |
| 5. $b = 39$ | 19. $y = 14$ | 33. $p = 48$ |
| 6. $w = 13$ | 20. $b = 21$ | 34. $a = 10$ |
| 7. $w = 41$ | 21. $k = 29$ | 35. $p = 35$ |
| 8. $t = 11$ | 22. $y = 25$ | 36. $a = 56$ |
| 9. $w = 25$ | 23. $m = 25$ | 37. $n = 5$ |
| 10. $k = 22$ | 24. $a = 21$ | 38. $x = 56$ |
| 11. $t = 14$ | 25. $k = 11$ | 39. $x = 24$ |
| 12. $n = 10$ | 26. $t = 5$ | 40. $k = 11$ |
| 13. $t = 23$ | 27. $p = 18$ | |
| 14. $n = 29$ | 28. $y = 9$ | |

Skill drill 2 (Medium basics)

- | | | |
|-------------|-------------|--------------|
| 1. $n = 6$ | 3. $y = -3$ | 5. $w = -9$ |
| 2. $t = -8$ | 4. $x = 8$ | 6. $m = -11$ |

$$\begin{aligned} 7. n &= -4 \\ 8. t &= 8 \\ 9. t &= 11 \\ 10. y &= 1 \\ 11. m &= -15 \\ 12. p &= 1 \\ 13. m &= 3 \\ 14. n &= -14 \\ 15. y &= 8 \\ 16. x &= 12 \\ 17. w &= \frac{3}{10} = 0.3 \end{aligned}$$

$$\begin{aligned} 18. y &= \frac{20}{7} = 2.857 \\ 19. x &= \frac{16}{6} = 2.667 \\ 20. x &= \frac{1}{3} = 0.333 \\ 21. y &= \frac{16}{5} = 3.2 \\ 22. y &= \frac{19}{2} = 9.5 \\ 23. w &= \frac{7}{3} = 2.333 \\ 24. a &= \frac{35}{2} = 17.5 \\ 25. x &= \frac{17}{6} = 2.833 \end{aligned}$$

$$\begin{aligned} 26. x &= \frac{6}{9} = 0.667 \\ 27. m &= \frac{26}{3} = 8.667 \\ 28. w &= \frac{14}{10} = 1.4 \\ 29. p &= \frac{17}{10} = 1.7 \\ 30. w &= \frac{14}{8} = 1.75 \\ 31. x &= \frac{29}{8} = 3.625 \\ 32. b &= \frac{33}{9} = 3.667 \end{aligned}$$

$$\begin{aligned} 19. a &= 3 \\ 20. b &= 2 \\ 21. k &= 24 \\ 22. y &= 10 \\ 23. a &= 42 \\ 24. m &= 24 \\ 25. y &= 30 \\ 26. a &= 16 \\ 27. x &= 18 \\ 28. t &= 3 \\ 29. b &= 18 \\ 30. k &= 24 \end{aligned}$$

$$\begin{aligned} 31. w &= 9 \\ 32. k &= 18 \\ 33. p &= 10 \\ 34. p &= 3 \\ 35. n &= 16 \\ 36. y &= 48 \end{aligned}$$

Extension

- $n = 4$
- $a = 15$

Exit ticket

- Level 1: $n = 22$
- Level 2: $x = 7$
- Level 3: $t = -24$

5.12 Solving two-step equations (booklet p.

15)

WE DO examples (teacher)

- Ex 1a: $t = 2$
- Ex 1b: $y = 3$
- Ex 2a: $m = 68$
- Ex 2b: $x = 9$
- Ex 3a: $m = 10$
- Ex 3b: $x = -4$

Easy · Set A (quick drill)

- | | | |
|------|------|--------|
| 1. 3 | 5. 5 | 9. 4 |
| 2. 2 | 6. 5 | 10. 3 |
| 3. 4 | 7. 4 | 11. 8 |
| 4. 3 | 8. 5 | 12. 18 |

Easy · Set B

- 68
- 9
- 12
- 20
- 12
- 10

Medium · Set C

- $m = 10$
- $k = 7$
- $x = -4$
- $x = -3$
- $d = -3$
- $x = 6$

Challenging · Set D

- $x = 12$
- $y = 9.5$
- $t = 4$
- $n = 7$

Skill drill 1 (Easy basics)

- | | | |
|-------------|--------------|--------------|
| 1. $m = 8$ | 7. $b = 6$ | 13. $n = 6$ |
| 2. $w = 4$ | 8. $n = 1$ | 14. $t = 3$ |
| 3. $b = 11$ | 9. $y = 6$ | 15. $x = 10$ |
| 4. $a = 5$ | 10. $x = 10$ | 16. $k = 9$ |
| 5. $k = 10$ | 11. $p = 11$ | 17. $y = 6$ |
| 6. $k = 8$ | 12. $y = 4$ | 18. $t = 6$ |

Skill drill 2 (Medium basics)

- | | | |
|--------------|--------------|--------------|
| 1. $n = -4$ | 11. $k = -8$ | 21. $x = 6$ |
| 2. $y = -4$ | 12. $w = 2$ | 22. $t = 11$ |
| 3. $p = 7$ | 13. $b = -9$ | 23. $k = 2$ |
| 4. $t = -8$ | 14. $w = -3$ | 24. $p = 9$ |
| 5. $p = -1$ | 15. $w = -1$ | 25. $b = 11$ |
| 6. $b = -3$ | 16. $a = -9$ | 26. $n = 11$ |
| 7. $k = 6$ | 17. $w = 1$ | 27. $a = 1$ |
| 8. $t = 9$ | 18. $y = 3$ | 28. $n = 7$ |
| 9. $t = -1$ | 19. $a = 12$ | |
| 10. $y = -4$ | 20. $m = 8$ | |

Extension

- $a = 4$
- $x = 4$

Exit ticket

- Level 1: $x = 6$
- Level 2: $n = 18$
- Level 3: $k = 9$

5.13 Equations and problem solving

(booklet p. 21)

WE DO examples (teacher)

- Ex 1a: C: $17N = 1003$
- Ex 1b: $N = 59$
- Ex 2a: $5N = 180$, $N = 36$ tickets
- Ex 2b: $5N - 180 = 2100$, $N = 456$ tickets
- Ex 3a: $4s = 52$, $s = 13$ cm
- Ex 3b: $n + n + 1 + n + 2 = 72$: 23, 24, 25

Easy · Set A (quick drill)

- $n + 8 = 20$
- $5n = 45$
- $n - 7 = 3$
- $n/2 = 16$
- $2n + 1 = 15$
- $3n - 4 = 11$
- $n + 11 = 30$, $n = 19$
- $6n = 42$, $n = 7$
- $n - 8 = 12$, $n = 20$
- $n/5 = 9$, $n = 45$

Easy · Set B

- 12
- 9
- 10
- 32
- 7
- 5

Medium · Set C

- \$2.50
- 13 cm
- \$138
- 12
- 13 km
- 5 tickets

Challenging · Set D

- 23, 24, 25

2. length 16 m, width 8 m
3. 456 tickets
4. \$80

Skill drill 1 (Easy basics)

- | | | |
|--------------------------------|-----------------------------------|-----------------------------------|
| 1. $2n + 8 = 36$, $n =$
14 | 8. $n + 9 = 26$, $n =$
17 | 15. $2n + 8 = 46$, $n =$
19 |
| 2. $7n = 98$, $n = 14$ | 9. $2n + 8 = 34$, $n =$
13 | 16. $n \div 4 = 16$, $n =$
64 |
| 3. $2n = 30$, $n = 15$ | 10. $8n = 56$, $n = 7$ | 17. $2n + 2 = 36$ |
| 4. $n + 2 = 18$, $n =$
16 | 11. $n \div 2 = 16$, $n =$
32 | 18. $2n + 5 = 27$ |
| 5. $2n + 9 = 41$, $n =$
16 | 12. $n \div 4 = 6$, $n = 24$ | 19. $4n = 12$ |
| 6. $8n = 152$, $n = 19$ | 13. $7n = 84$, $n = 12$ | 20. $2n + 6 = 16$ |
| 7. $2n + 6 = 40$, $n =$
17 | 14. $n - 6 = 15$, $n =$
21 | 21. $n \div 6 = 20$ |
| | | 22. $2n + 4 = 44$ |
| | | 23. $3n - 9 = 51$ |
| | | 24. $n - 8 = 11$ |

Skill drill 2 (Medium basics)

- | | | |
|--------------------------------|-------------------------------------|--------------------------------------|
| 1. $10n = 160$, 16
pencils | 5. $14n = 168$, 12
pencils | 9. $42 + 40 + x =$
180, $x = 98$ |
| 2. $4s = 36$, 9 cm | 6. $4s = 28$, 7 cm | 10. $37 + 58 + x =$
180, $x = 85$ |
| 3. $4s = 16$, 4 cm | 7. $49 + 32 + x =$
180, $x = 99$ | 11. $35 + 56 + x =$
180, $x = 89$ |
| 4. $9n = 81$, 9
pencils | 8. $50 + 34 + x =$
180, $x = 96$ | 12. $61 + 68 + x =$
180, $x = 51$ |

Extension

1. 7.5 weeks
2. 32 and 18

Exit ticket

1. Level 1: $n + 14 = 31$, $n = 17$
2. Level 2: $4b = 62$, $b = \$15.50$
3. Level 3: $2n + 9 = 41$, $n = 16$